



REAL AND NOMINAL GDP

PART A

There are two ways to calculate **GROSS DOMESTIC PRODUCT**. Answer the following questions regarding **REAL** and **NOMINAL GDP**.

Define **NOMINAL** GROSS DOMESTIC PRODUCT.

The dollar value of all final goods and services produced within a country's borders in one year, expressed in current dollar value (unadjusted for inflation).

Define **REAL** GROSS DOMESTIC PRODUCT.

The dollar value of all final goods and services produced within a country's borders in one year, expressed in constant dollar value (adjusted for inflation).

Define the **STANDARD OF LIVING**.

The quality of life in a society, measured in part by how well an economy is doing, taking into account a nation's population size.

Finish this sentence to state an economic law:

**"A NATION'S STANDARD OF LIVING IS BEST MEASURED BY CHANGES IN ITS
REAL GDP PER CAPITA"**

Explain what **REAL GDP PER CAPITA** identifies in a society.

It identifies the quantity of goods and services attributed and made available for each member of society. In short, Real GDP per person in the population.

Name **THREE** factors that lead some nations to have higher **REAL** GDPs than others.

AVAILABILITY AND
QUANTITY OF NATURAL
RESOURCES

AVAILABILITY OF CAPITAL

ECONOMIC SYSTEMS

CALCULATING

REAL GDP

PART B

First, **CALCULATE THE NOMINAL GDP** for each country in 2015 and 2016.
Then, calculate the **REAL GDP** in 2016. Was there **GDP GROWTH**?

UNITED STATES GDP DATA - 2015

	Steel	Wheat	Corn	Sugar
PRICE	\$10	\$2	\$3	\$1
QUANTITY	400,000	100,000	200,000	200,000
TOTAL	\$4,000,000	\$200,000	\$600,000	\$200,000

UNITED STATES GDP DATA - 2016

	Steel	Wheat	Corn	Sugar
PRICE	\$10	\$3	\$2	\$2
QUANTITY	450,000	100,000	300,000	300,000
TOTAL	\$4,500,000	\$300,000	\$600,000	\$600,000

RUSSIA GDP DATA - 2015

	Copper	Salt	Lumber	Oil
PRICE	\$20	\$5	\$10	\$40
QUANTITY	200,000	200,000	300,000	50,000
TOTAL	\$4,000,000	\$1,000,000	\$3,000,000	\$2,000,000

RUSSIA GDP DATA - 2016

	Copper	Salt	Lumber	Oil
PRICE	\$12	\$6	\$20	\$20
QUANTITY	250,000	150,000	400,000	50,000
TOTAL	\$3,000,000	\$900,000	\$8,000,000	\$1,000,000

REMEMBER!

$$\text{REAL GDP IN 2016} = \frac{2016 \text{ QUANTITY}}{2015 \text{ PRICES}} \times 2015 \text{ QUANTITY}$$



2015 Nominal GDP	\$5,000,000
2016 Nominal GDP	\$6,000,000
2016 Real GDP	\$5,900,000

EXPANSION or CONTRACTION



2015 Nominal GDP	\$10,000,000
2016 Nominal GDP	\$12,900,000
2016 Real GDP	\$11,750,000

EXPANSION or CONTRACTION

CALCULATING

REAL GDP

GERMANY GDP DATA - 2015

	Iron	Coal	Wheat	Granite
PRICE	\$15	\$5	\$2	\$20
QUANTITY	150,000	500,000	250,000	100,000
TOTAL	\$2,250,000	\$2,500,000	\$500,000	\$2,000,000

GERMANY GDP DATA - 2016

	Iron	Coal	Wheat	Granite
PRICE	\$20	\$8	\$3	\$25
QUANTITY	100,000	500,000	300,000	100,000
TOTAL	\$2,000,000	\$4,000,000	\$900,000	\$2,500,000

JAPAN GDP DATA - 2015

	TVs	Cameras	Silk	Fish
PRICE	\$1000	\$500	\$50	\$2
QUANTITY	2000	10,000	200,000	500,000
TOTAL	\$2,000,000	\$5,000,000	\$10,000,000	\$1,000,000

JAPAN GDP DATA - 2016

	TVs	Cameras	Silk	Fish
PRICE	\$1500	\$400	\$55	\$1
QUANTITY	2500	15,000	250,000	750,000
TOTAL	\$3,750,000	\$6,000,000	\$13,750,000	\$750,000

REMEMBER!

$$\text{REAL GDP IN 2016} = \frac{2016 \text{ QUANTITY}}{2015 \text{ PRICES}} \times$$



2015 Nominal GDP \$7,250,000

2016 Nominal GDP \$9,400,000

2016 Real GDP \$6,600,000

EXPANSION or **CONTRACTION**



2015 Nominal GDP \$18,000,000

2016 Nominal GDP \$24,250,000

2016 Real GDP \$24,000,000

EXPANSION or CONTRACTION

CALCULATING

REAL GDP

PART C

REAL GDP PER CAPITA is the most accurate measurement of a nation's **STANDARD OF LIVING**. Calculate the **REAL GDP PER CAPITA** for each country in 2015 and 2016. Then indicate what happened to the **STANDARD OF LIVING** in each country.

REMEMBER!

$$\text{REAL GDP PER CAPITA} = \frac{\text{REAL GDP}}{\text{POPULATION}}$$

UNITED STATES



	2015	2016
REAL GDP	\$12 TRILLION	\$15 TRILLION
POPULATION	300 MILLION	300 MILLION

2015 Real GDP per capita \$40,000

2016 Real GDP per capita \$50,000

Did the standard of living increase, decrease or stay the same? Explain.

Increased. Real GDP per capita has increased from \$40,000 to \$50,000, and each person has \$10,000 more goods and services in 2015 than they had in 2016.

GERMANY



	2015	2016
REAL GDP	\$3 TRILLION	\$5 TRILLION
POPULATION	120 MILLION	200 MILLION

2015 Real GDP per capita \$25,000

2016 Real GDP per capita \$25,000

Did the standard of living increase, decrease or stay the same? Explain.

Stayed the same. Real GDP per capita remained stagnant at \$25,000, so each person had the same amount of goods and services in 2015 than they had in 2016.

JAPAN



	2015	2016
REAL GDP	\$5 TRILLION	\$6 TRILLION
POPULATION	10 MILLION	30 MILLION

2015 Real GDP per capita \$500,000

2016 Real GDP per capita \$200,000

Did the standard of living increase, decrease or stay the same? Explain.

Decreased. Real GDP per capita has decreased from \$500,000 to \$200,000, and each person has less goods and services in 2015 than they had in 2016.